

Esercizi

$$1) E_x = \int_{-\infty}^{+\infty} |x(t)|^2 dt = \int_{-\infty}^{+\infty} 25 t^2 e^{-\frac{t^2}{2}} dt$$

$$= 25 \int_{-\infty}^{+\infty} t^2 e^{-\frac{t^2}{2}} dt = (\text{per parti})$$

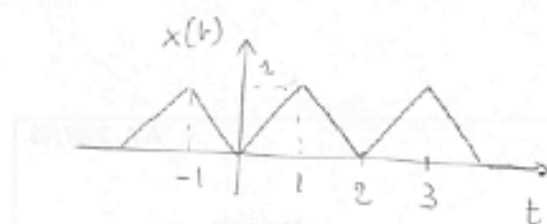
$$= 25 \left[-e^{-\frac{t^2}{2}} \cdot t \right]_{-\infty}^{+\infty} + \int_{-\infty}^{+\infty} e^{-\frac{t^2}{2}} dt = 25 \int_{-\infty}^{+\infty} e^{-\frac{t^2}{2}} dt$$

$$= 25 \sqrt{2\pi}$$

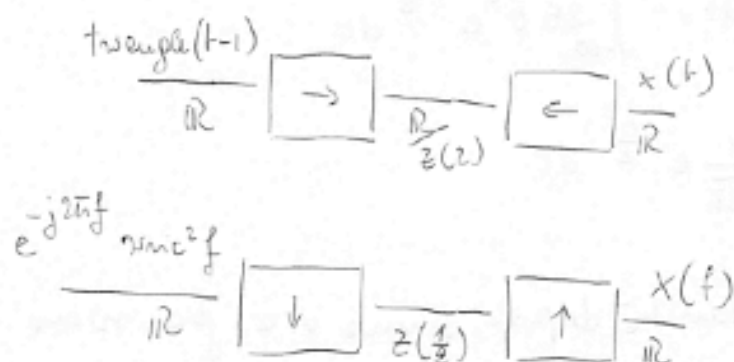
$$\int_{-\infty}^{+\infty} e^{-\frac{t^2}{2}} dt = \int_{-\infty}^{+\infty} e^{-\pi \left(\frac{t}{\sqrt{2\pi}} \right)^2} dt = \sqrt{2\pi} e^{-\pi \left(\sqrt{2\pi} f \right)^2} \Big|_{f=0}$$

$$= \sqrt{2\pi}$$

2)



$$x(t) = \text{rep}_2 \text{triangle}(t-1), \quad t \in \mathbb{R}$$



$$\begin{aligned} X(f) &= \frac{1}{2} \sum_{k=-\infty}^{+\infty} e^{-j2\pi \frac{k}{2}} \text{sinc}^2\left(\frac{k}{2}\right) \delta_{\mathbb{R}}\left(f - \frac{k}{2}\right) \\ &= \frac{1}{2} \sum_{k=-\infty}^{+\infty} (-1)^k \text{sinc}^2\left(\frac{k}{2}\right) \delta_{\mathbb{R}}\left(f - \frac{k}{2}\right) \end{aligned}$$

3) Le f_f di Fourier di $\text{rect}\left(\frac{kT}{5T}\right)$ è $5T \text{sinc}_5(f5T)$.

$$\text{Dunque, } H(f) = \int_{\mathbb{R}} e^{j2\pi kT \frac{f}{2T}} \text{rect}\left(\frac{kT}{5T}\right) = 5T \text{sinc}_5\left(5T\left(f - \frac{1}{2T}\right)\right)$$

Dato che gli esponenziali complessi sono autofunzioni del filtro, l'uscita $y(kT)$ all'ingresso $x(kT) = e^{+j2\pi kT \frac{f}{2T}}$ risulta

$$y(kT) = x(kT) \cdot H\left(\frac{1}{2T}\right) = 5T (-1)^k$$

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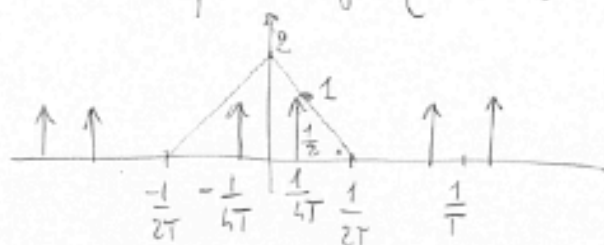
$$4) \quad \frac{x(kT)}{z(T)} \boxed{\uparrow} \xrightarrow{R} \boxed{L(t)} \xrightarrow{1/R} \frac{y(t)}{1/R}$$

$$\frac{X(f)}{R/z(\frac{1}{T})} \boxed{\leftarrow} \xrightarrow{R} \boxed{H(f)} \xrightarrow{1/R} \frac{Y(f)}{1/R}$$

$$x(kT) = \cos 2\pi kT f_0 \quad f_0 = \frac{1}{4T}$$

$$X(f) = \frac{\delta_{R/z(\frac{1}{T})}(f-f_0) + \delta_{R/z(\frac{1}{T})}(f+f_0)}{2}$$

$$H(f) = \frac{2T}{T} \text{triangola} \left(\frac{f \cdot 2T}{1} \right)$$



$$Y(f) = \frac{\delta_{R/z(\frac{1}{T})}(f-\frac{1}{4T}) + \delta_{R/z(\frac{1}{T})}(f+\frac{1}{4T})}{2}$$

$$y(t) = \cos 2\pi \frac{1}{4T} \cdot t \quad t \in \mathbb{R}$$

$$\begin{aligned}
 5) \quad h(f) &= \frac{T}{(1 - 0.1e^{-j2\pi fT})(1 - 0.3e^{-j2\pi fT})} \\
 &= \frac{-\frac{1}{2}T}{1 - 0.1e^{-j2\pi fT}} + \frac{\frac{3}{2}T}{(1 - 0.3e^{-j2\pi fT})}
 \end{aligned}$$

$$e_c(kT) = -\frac{1}{2}(0.1)^k 1_0(kT) + \frac{3}{2}(0.3)^k 1_0(kT)$$

Esercizi

$$1) E_x = \int_{-\infty}^{+\infty} |x(t)|^2 dt = \int_{-\infty}^{+\infty} 25 t^2 e^{-\frac{t^2}{2}} dt$$

$$= 25 \sqrt{2\pi} \int_{-\infty}^{+\infty} t^2 \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

$\frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}}$ è la densità di prob. di una v.e. gaussiana
e media nulla e varianza $\sigma^2 = 1$. Dunque

$$\int_{-\infty}^{+\infty} t^2 \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt = \sigma^2 = 1$$

$$E_x = 25 \sqrt{2\pi}$$

In alternativa l'esercizio si risolve, posto

$$s(t) = 25 t^2 e^{-\frac{t^2}{2}}$$

calcolando

$$\int_{-\infty}^{+\infty} s(t) dt = S(0)$$

La t.f. di Fourier di $s(t)$ si calcola a partire da
quella di $y(t) = e^{-\frac{t^2}{2}}$ e sfruttando due volte le regole

$$-j\omega t y(t) \leftrightarrow \frac{dY(f)}{df}$$